Automation Testing Practice - Test Plan

Project Overview:

This project aims to automate the testing of the Automation Testing Practice application using Selenium WebDriver, Java, TestNG, Maven, Jenkins, and GitHub. The automation framework is developed using the Page Object Model (POM) design pattern.

Objective:

The objective of this project is to automate key functionalities of the Automation Testing Practice Website, including form validation, alerts, frames, windows, web tables, and link verification, thereby reducing manual testing effort, improving test coverage, and ensuring application quality through efficient automated testing.

Modules Covered:

- GUI Elements
- Upload Files
- Pagination Web Table
- Account Overview
- Forms
- Tabs
- Dynamic button
- Alerts and Popups
- Mouse Hover
- Double Click
- Drag and Drop
- Slider
- Scrolling Dropdown
- Laptop Links

- Broken Links

Tools used:

- Java
- Selenium WebDriver
- TestNG
- Maven
- Jenkins
- GitHub

Test Approach:

The framework follows the Page Object Model (POM) design pattern. Test scripts are created for positive test scenarios and known defect scenarios are maintained separately.

Test Environment:

- Operating System: Windows
- Browser: Chrome, Firefox
- Application URL: https://testautomationpractice.blogspot.com/

Test Scenarios:

Form Submission:

- Submit the form with valid input data.
- Verify successful form submission message.

Radio Buttons:

- Select different radio button options.
- Verify only one option can be selected at a time.

Checkboxes:

- Select and deselect checkboxes.
- Verify multiple checkboxes can be selected simultaneously.

Dropdown Handling:

- Select values from single-select dropdowns.
- Verify the selected value is displayed correctly.

Alert Handling:

- Trigger simple, confirmation, and prompt alerts.
- Verify alert messages and user actions.

Multiple Windows/Tabs:

- Open a new browser window/tab.
- Switch between windows and verify page content.

Web Table Validation:

- Retrieve and validate data from web tables.
- Verify row and column values.

Mouse Actions:

- Perform hover actions on menu items.
- Execute drag-and-drop, right-click, and double-click operations.

File Upload:

- Upload a valid file successfully.
- Verify upload confirmation message.

Broken Links:

- Verify all hyperlinks on the website are functional and do not return broken link errors.
- Validate that each link redirects to the correct webpage successfully.

Laptop Links:

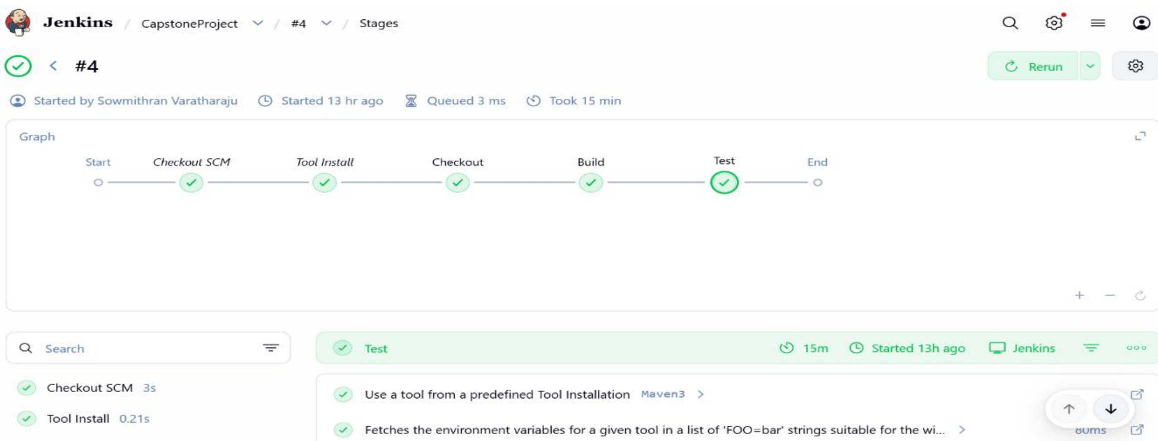
- Verify the Laptop link navigates to the Laptop products page successfully.
- Validate that laptop product details are displayed correctly after navigation.

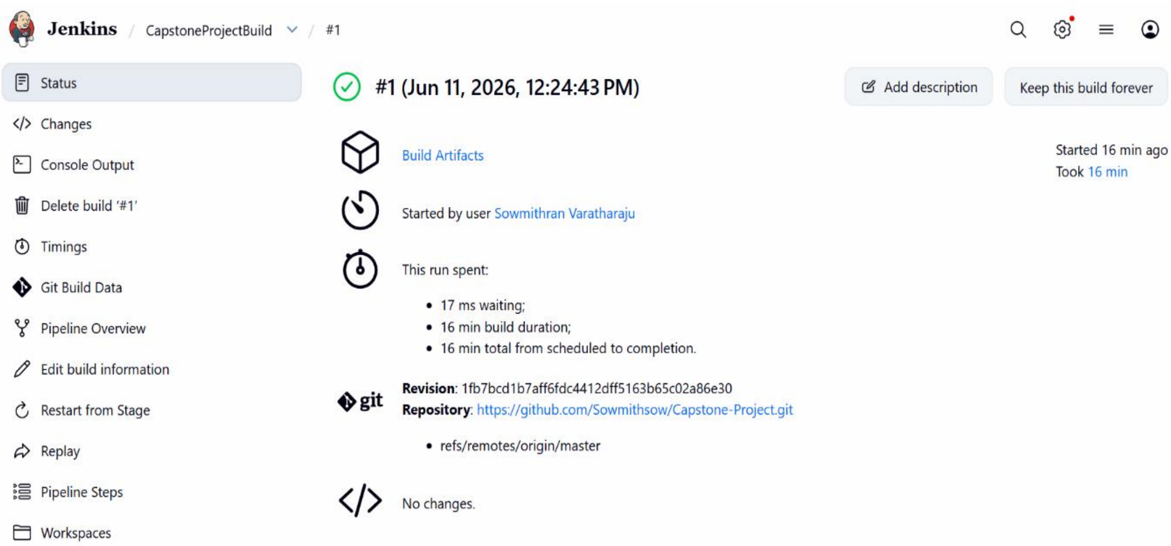
Defects:

- Name Field accept Numeric Values without Validation.
- Name Field accept Special Characters without Validation.
- Name Field accept Alphanumeric Characters without Validation.
- Email field accepts invalid email format without Validation.
- Email field accepts invalid domain format without Validation.
- Mobile Number field accept lessthan 10 digits without Validation.
- Mobile Number field accept alphanumeric characters without Validation.
- Mobile Number field accept special characters without Validation.
- Mobile Number field accept morethan 10 digits without Validation.
- Address field accept insufficient address details without Validation.
- Address field accept numeric only input without Validation.
- Address field accept special characters only without Validation.

JENKINS INTEGRATION:

The automation framework is integrated with Jenkins to enable continuous and controlled test execution. The Jenkins pipeline is configured to automatically fetch the latest code from the GitHub repository, perform a Maven build to resolve dependencies and compile the project, and execute the TestNG test suite. The execution is initiated manually using the “Build Now” option, and the test results along with execution logs are monitored through the Jenkins console output for effective analysis and debugging.





Conclusion:

The Automation Testing Practice Framework successfully automates major banking functionalities. The framework supports automated execution, reporting, cross-browser testing, and Jenkins integration.

DEFECT REPORT

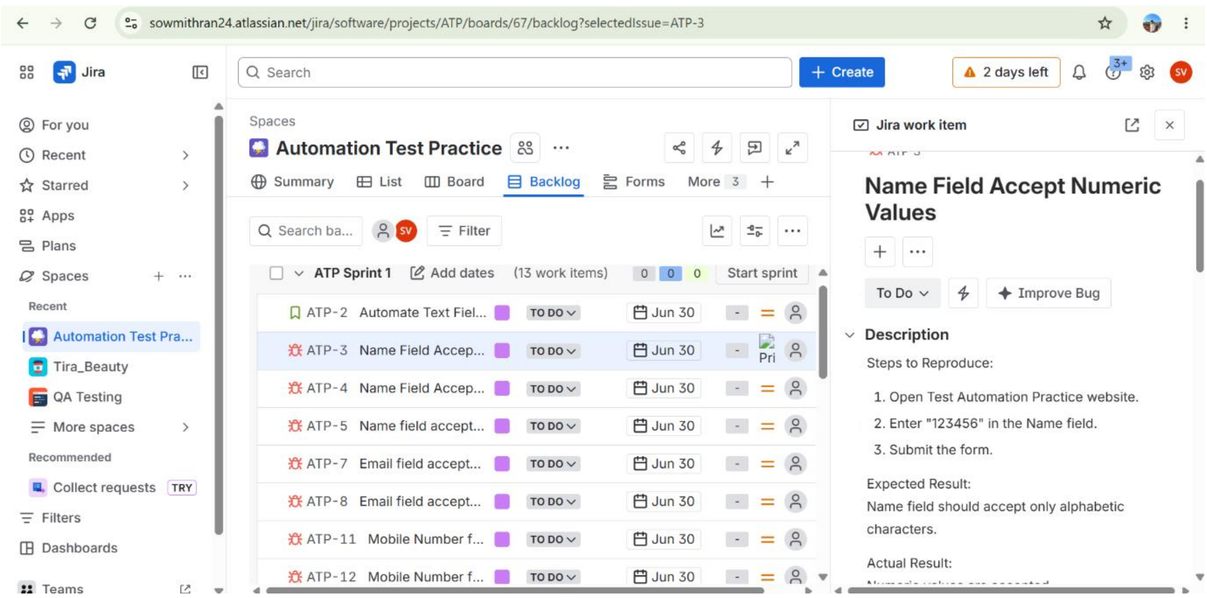
Bug_01: Name Field accept Numeric Values without Validation

Steps:

1. Open Test Automation Practice website.
2. Enter "123456" in the Name field.
3. Submit the form.

Expected Result:
Name field should accept only alphabetic characters.

Actual Result:
Numeric values are accepted



Bug_02: Name Field accept Special Characters without Validation

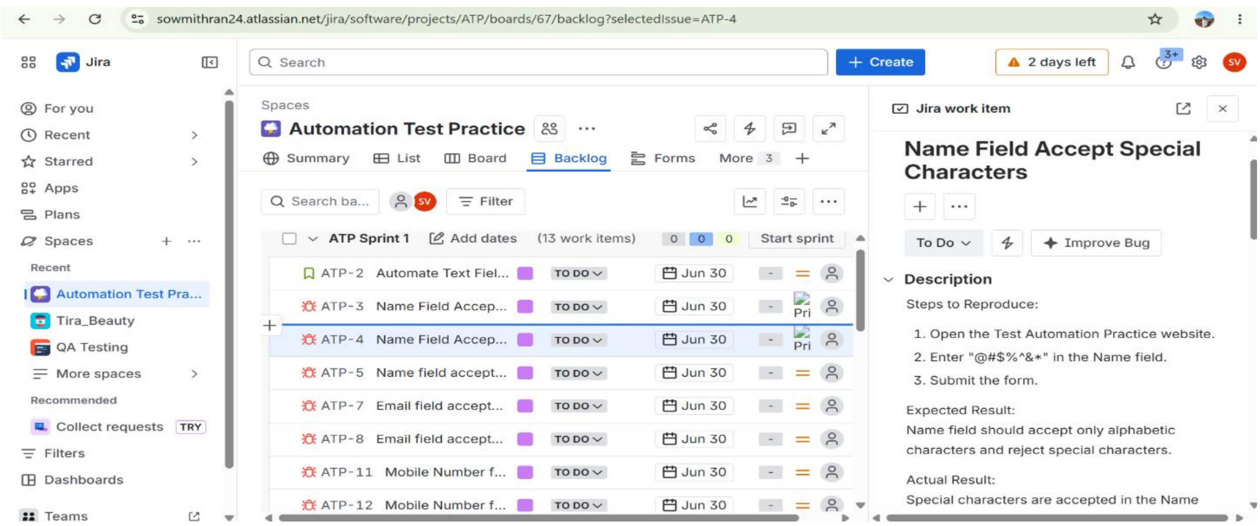
<u>Steps</u>:

1. Open the Test Automation Practice website.
2. Enter "@#$%^&*" in the Name field.

3.Submit the form.

<u>Expected Result</u>:
Name field should accept only alphabetic characters and reject special characters.

<u>Actual Result</u>:
Special characters are accepted in the Name field without any validation message.

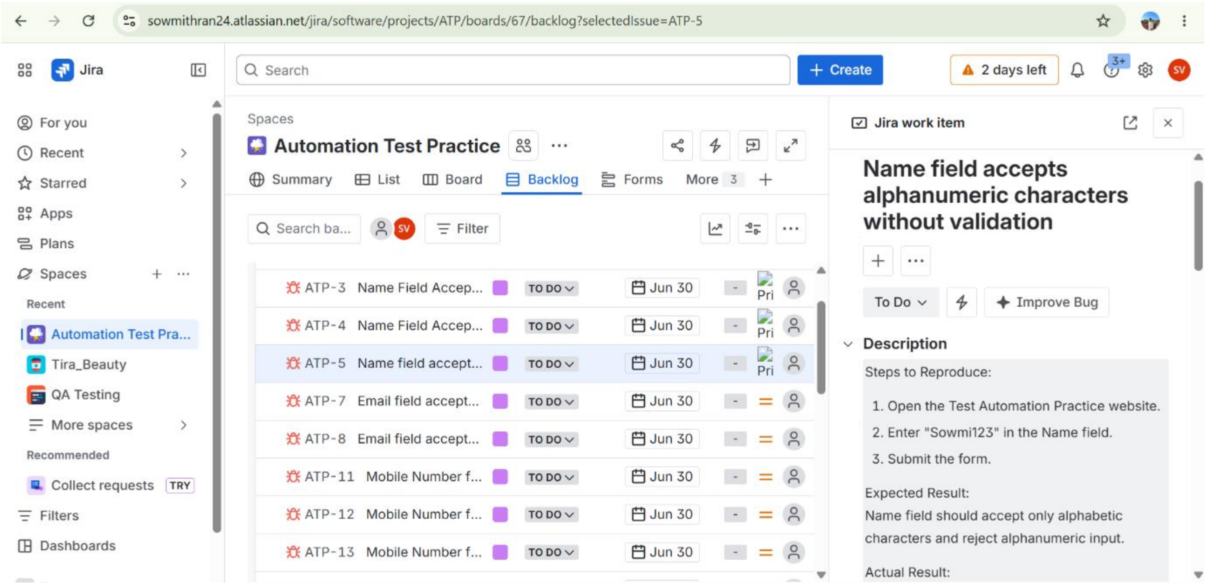


Bug_03: Name Field accept Alphanumeric Characters without Validation

1. Open the Test Automation Practice website.
2. Enter "Sowmi123" in the Name field.
3. Submit the form.

<u>Expected Result</u>:
Name field should accept only alphabetic characters and reject alphanumeric input.

Actual Result:
Alphanumeric characters are accepted in the Name field without any validation message.



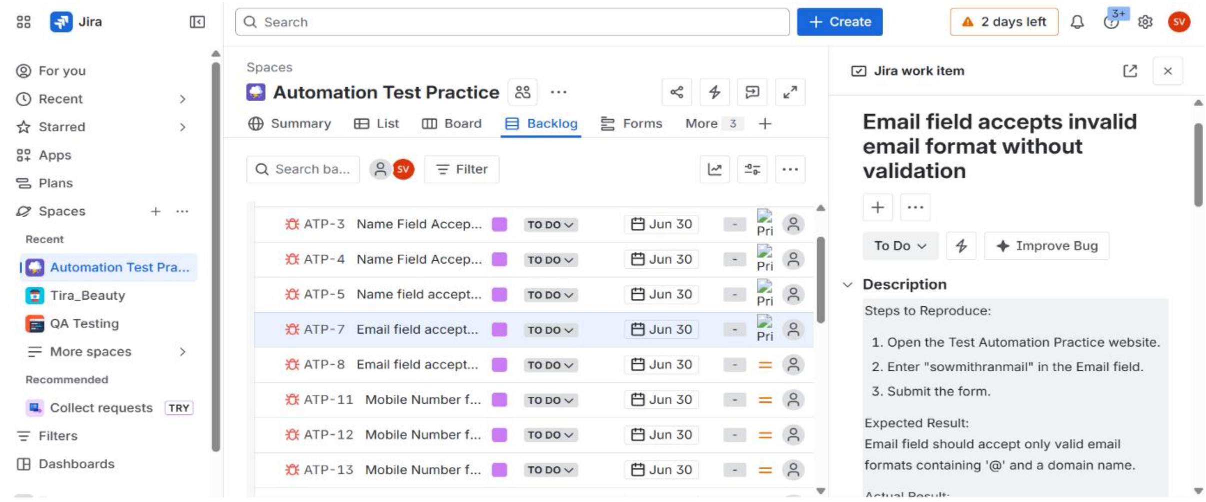
Bug_04: Email field accepts invalid email format without Validation

Steps:

1. Open the Test Automation Practice website.
2. Enter "sowmithranmail" in the Email field.
3. Submit the form.

Expected Result:
Email field should accept only valid email formats containing '@' and a domain name.

Actual Result:
The Email field accepts "sowmithranmail" without displaying any validation message.



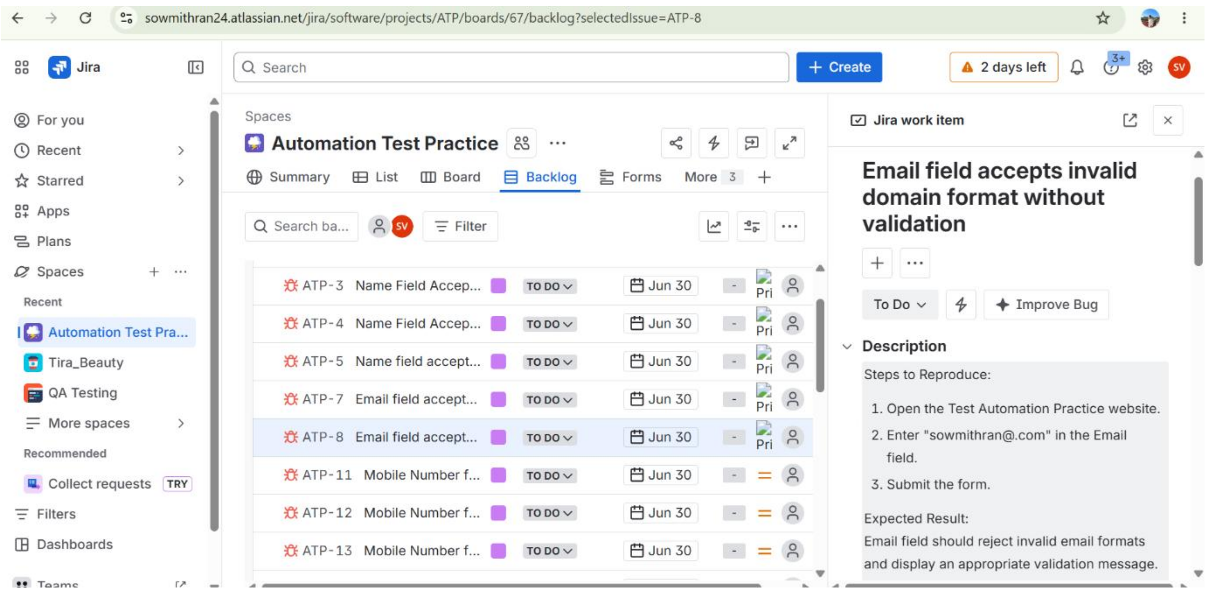
Bug_05: Email field accepts invalid domain format without Validation

Steps:

1. Open the Test Automation Practice website.
2. Enter "sowmithran@.com" in the Email field.
3. Submit the form.

Expected Result:
Email field should reject invalid email formats and display an appropriate validation message.

Actual Result:
The Email field accepts "sowmithran@.com" without displaying any validation message.



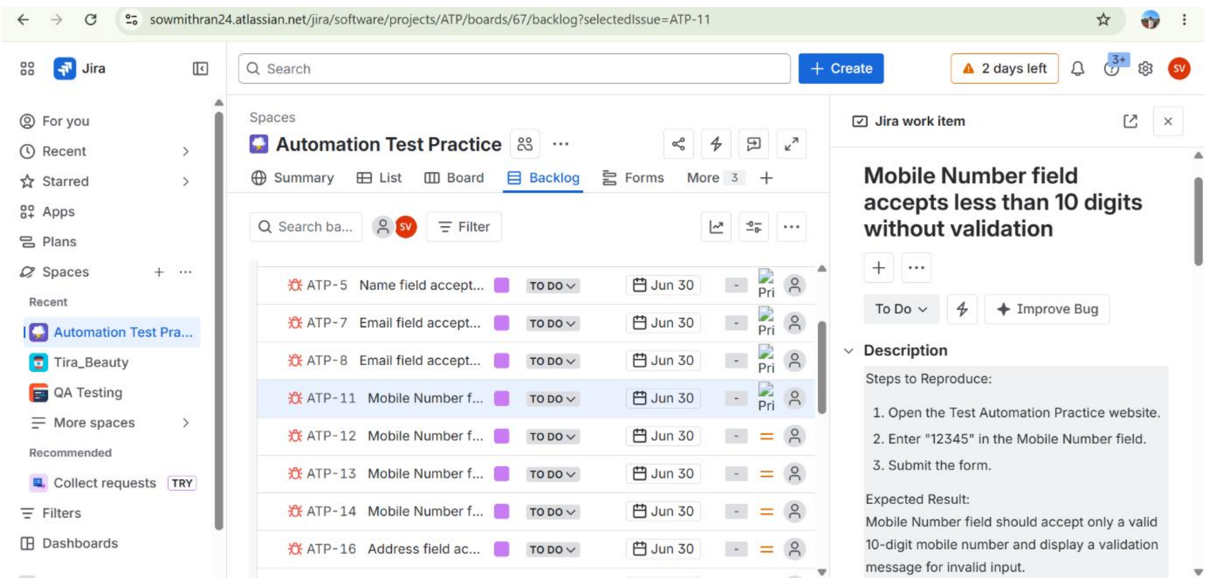
Bug_06: Mobile Number field accept lessthan 10 digits without Validation

Steps:

1. Open the Test Automation Practice website.
2. Enter "12345" in the Mobile Number field.
3. Submit the form.

Expected Result:
Mobile Number field should accept only a valid 10-digit mobile number and display a validation message for invalid input.

Actual Result:
The Mobile Number field accepts "12345" without displaying any validation message.



Bug_07: Mobile Number field accept alphanumeric characters without Validation

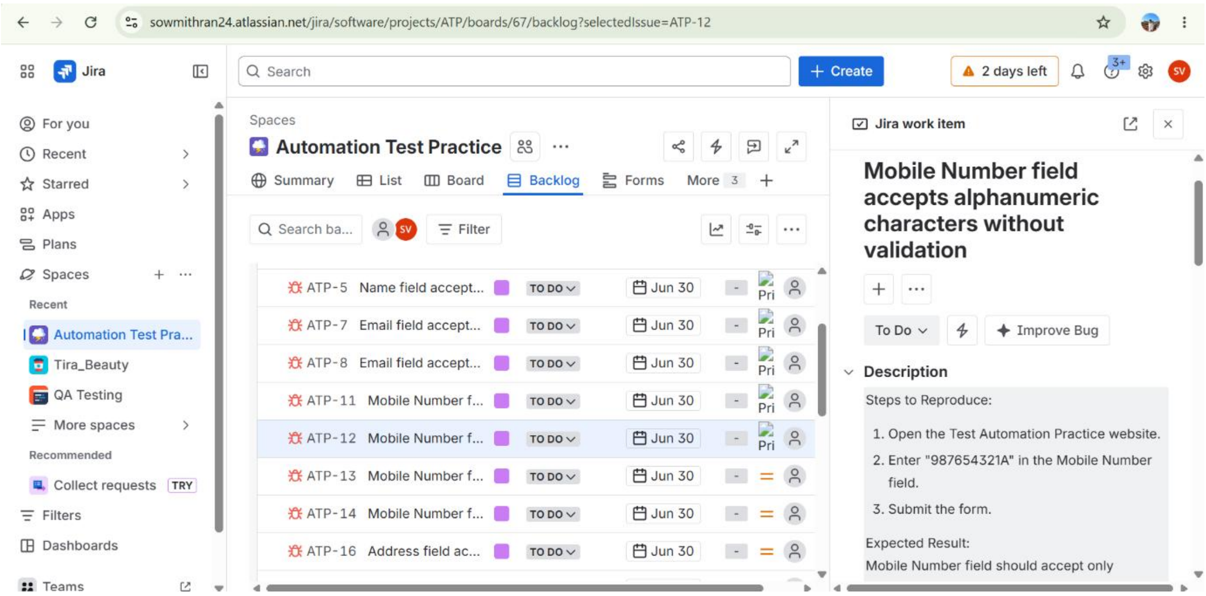
Steps:

1. Open the Test Automation Practice website.
2. Enter "987654321A" in the Mobile Number field.

3. Submit the form.

Expected Result:
Mobile Number field should accept only numeric values and reject alphanumeric characters.

Actual Result:
The Mobile Number field accepts "987654321A" without displaying any validation message.



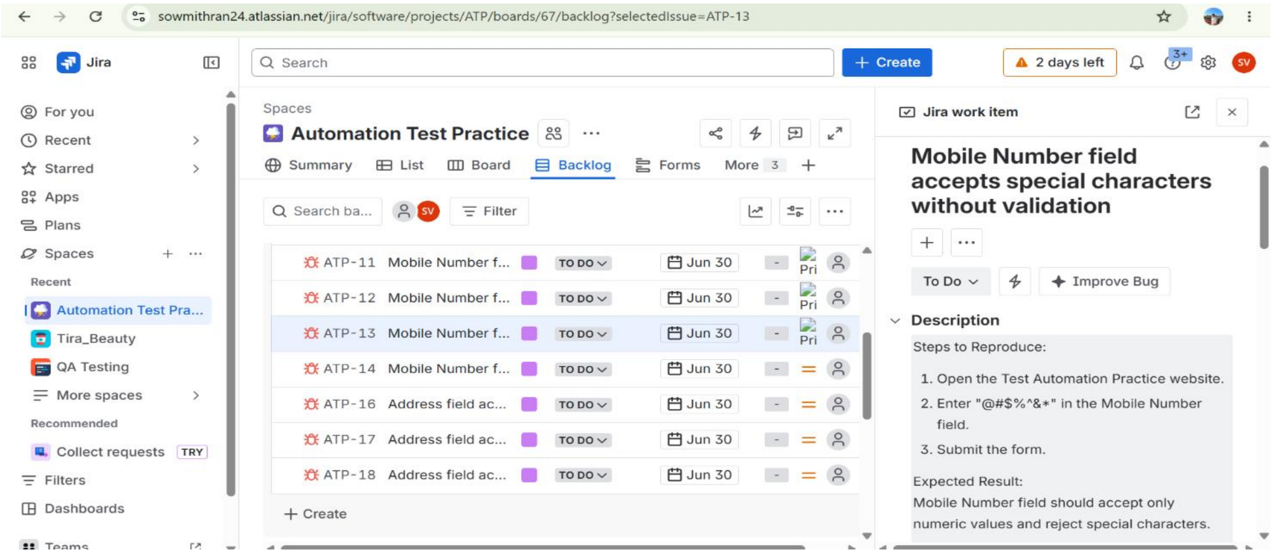
Bug_08: Mobile Number field accept special characters without Validation

Steps:

1. Open the Test Automation Practice website.
2. Enter "@#$%^&*" in the Mobile Number field.
3. Submit the form.

Expected Result:
Mobile Number field should accept only numeric values and reject special characters.

Actual Result:
The Mobile Number field accepts "@#$%^&*" without displaying any validation message.



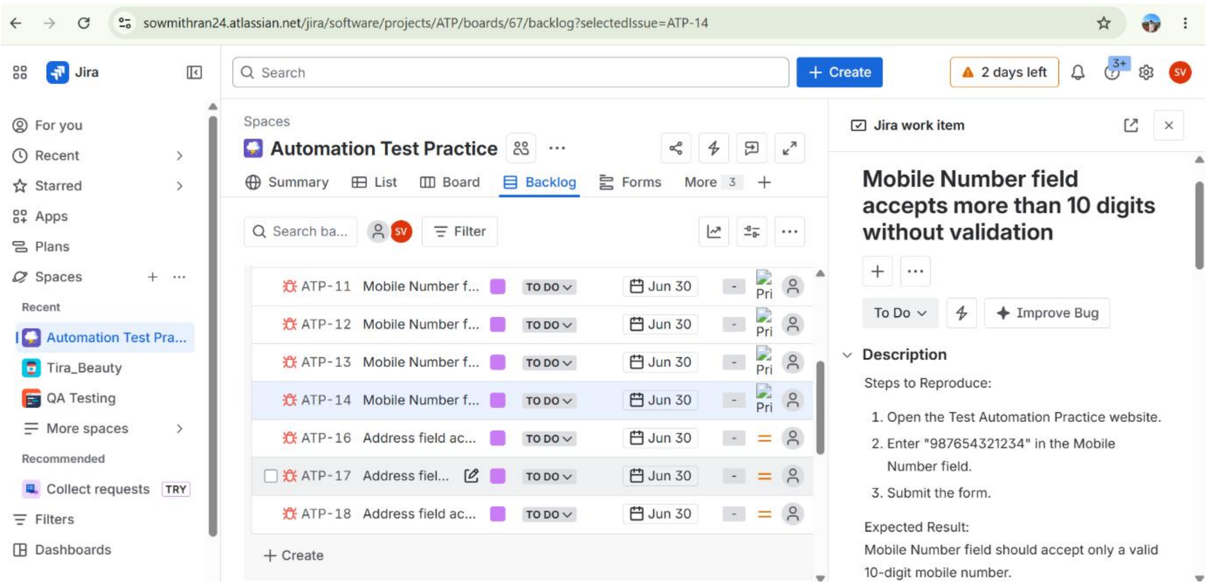
Bug_09: Mobile Number field accept morethan 10 digits without Validation

Steps:

1. Open the Test Automation Practice website.
2. Enter “987654321234” in the Mobile Number field.
3. Submit the form.

Expected Result:
Mobile Number field should accept only a valid 10-digit mobile number.

Actual Result:
The Mobile Number field accepts “987654321234” without displaying any validation message.



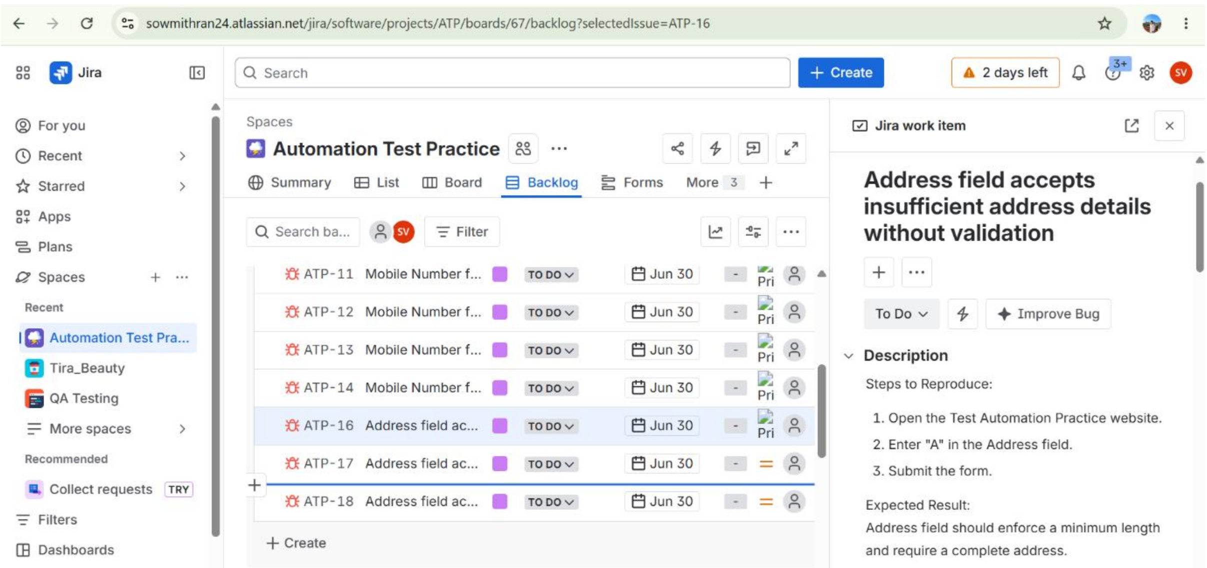
Bug_10: Address field accept insufficient address details without Validation

Steps:

1. Open the Test Automation Practice website.
2. Enter “A” in the Address field.
3. Submit the form.

Expected Result:
Address field should enforce a minimum length and require a complete address.

Actual Result:
The Address field accepts “A” without displaying any validation message.



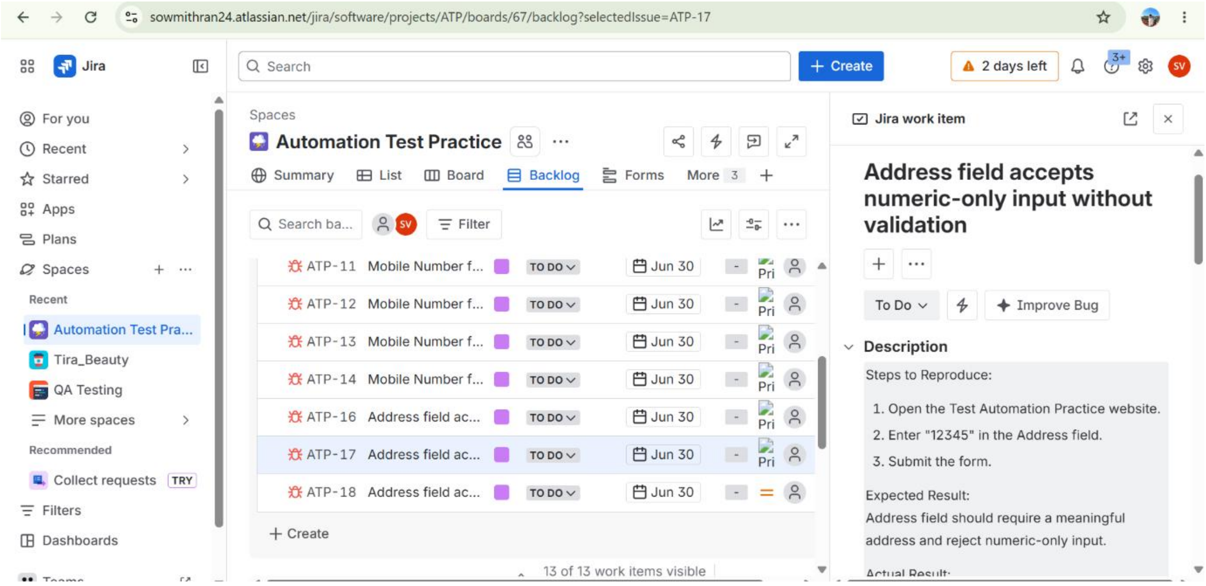
Bug_11: Address field accept numeric only input without Validation

Steps:

1. Open the Test Automation Practice website.
2. Enter “12345” in the Address field.
3. Submit the form.

Expected Result:
Address field should require a meaningful address and reject numeric-only input.

<u>Actual Result</u>:
The Address field accepts “12345” without displaying any validation message.



Bug_12: Address field accept special characters only without Validation

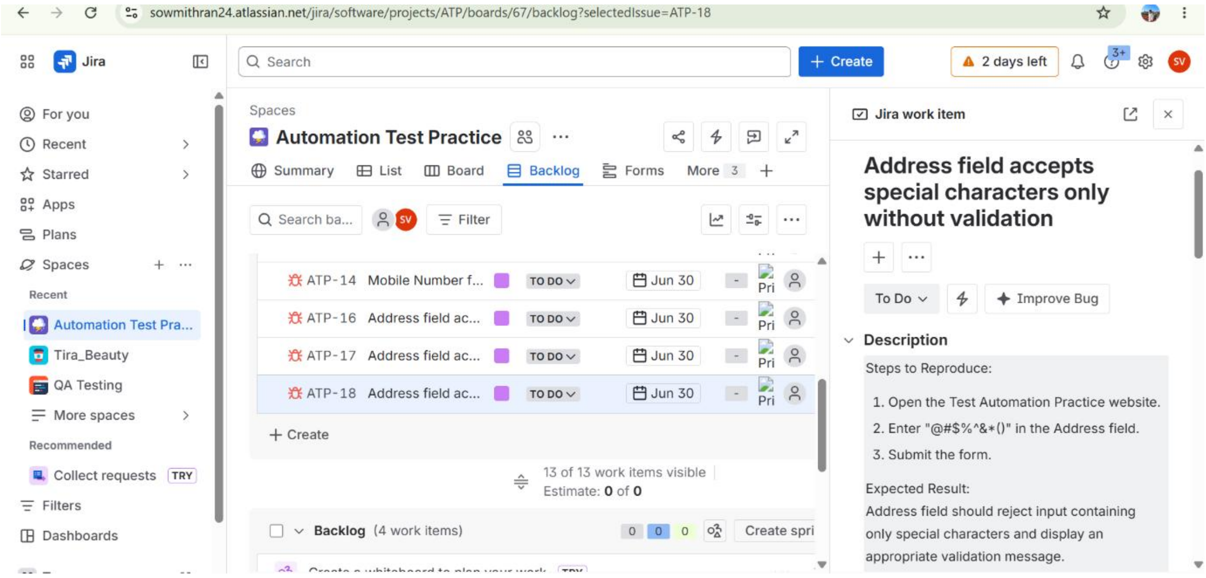
<u>Steps</u>:

1. Open the Test Automation Practice website.
2. Enter “@#$%^&*()” in the Address Field.
3. Submit the form.

<u>Expected Result:</u>
Address field should reject input containing only special characters and display an appropriate validation message.

<u>Actual Result</u>:

The Address field accepts “@#$%^&*()” without displaying any Validation message.



TESTCASE ANALYSIS REPORT

Overview:

The Automation Testing Practice Website Framework was executed to validate the core functionalities of the application, including user interactions, form submissions, alerts, windows, frames, web tables, and dropdown handling. Automated test cases were developed using Selenium WebDriver, Java, TestNG, Maven, and Jenkins to ensure functional accuracy, reliability, and efficient test execution through continuous integration.

Test Execution Summary :

Metric	Count
Total Test Cases	34
Executed Test Cases	34
Passed Test Cases	34

Failed Test Cases	0
Blocked Test Cases	0

Module-wise Analysis :

Module	**Status**
Form Submission	Passed with Defects Identified
Radio Button Handling	Passed
Checkbox Validation	Passed
Dropdown Selection	Passed
Alert Handling	Passed
Multiple Windows/Tabs	Passed
Web Table Validation	Passed
Mouse Actions (Hover, Drag & Drop, Double Click)	Passed
File Upload	Passed
Dynamic Button	Passed
Laptop Links	Passed
Broken Links	Passed

Defect Analysis:

Bug ID	Description	Severity
BUG-1	Name Field accept Numeric Values without Validation.	Medium

BUG-2	Name Field accept Special Characters without Validation	Medium
BUG-3	Name Field accept Alphanumeric Characters without Validation	Medium
BUG-4	Email field accepts invalid email format without Validation	High
BUG-5	Email field accepts invalid domain format without Validation	High
BUG-6	Mobile Number field accept lessthan 10 digits without Validation	High
BUG-7	Mobile Number field accept alphanumeric characters without Validation	High
BUG-8	Mobile Number field accept special characters without Validation	High
BUG-9	Mobile Number field accept morethan 10 digits without Validation	High
BUG-10	Address field accept insufficient address details without Validation	Low
BUG-11	Address field accept numeric only input without Validation	Medium
BUG-12	Address field accept special characters only without Validation	Medium

Test Coverage :

- User Interface Validation
- Form Handling
- Dropdowns
- Alerts and Pop-ups
- Browser Windows and Tabs
- Web Tables
- Mouse Actions
- Synchronization
- Reporting and CI/CD

Challenges Faced:

- Jenkins Pipeline Setup
- Selenium WebDriver Version Compatibility
- Synchronization Issues
- Managing Test Dependencies
- Handling Dynamic Elements

Conclusion:

The automation framework successfully validated the major functionalities of the Automation Testing Practice Website, including form handling, alerts, frames, windows, web tables, dropdowns, and mouse actions. The framework supports automated execution through Jenkins, utilizes TestNG for test management, and generates detailed execution reports using Extent Reports, ensuring efficient test monitoring and defect tracking.

PROJECT SUMMARY:

The Automation Testing Practice Website Automation Framework was developed using Selenium WebDriver, Java, TestNG, Maven, Jenkins, and GitHub following the Page Object Model (POM) design pattern. The framework automates key web functionalities including form validation, alerts handling, frames, windows, web tables, dropdowns, mouse actions, file upload, and link verification.

A total of 20+ automated test cases were executed. Most test cases passed successfully, while a few defects were identified during negative and boundary testing scenarios, particularly in input validation fields such as name, email, mobile number, and address. The identified issues highlight missing validation checks in the application.

Jenkins was integrated for continuous test execution, and Extent Reports were generated to provide detailed execution results and logs. The framework supports cross-browser testing and enhances test coverage, reduces manual effort, and improves overall testing efficiency through reusable automation components.